

FULL STACK WEB DEVELOPMENT

USE CASE – ASSIGNMENT

Subject Code: 21CSE354T

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PROGRAM / SECTION	B. Tech CSE - A
YEAR / SEMESTER	III / V
ASSIGNMENT / USE CASE NO.	3
DATE OF SUBMISSION	

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ACADEMIC YEAR	2026 – 2027

“Design • Develop • Deploy”

Password Strength Checker

1. TITLE

A Dynamic Password Strength Indicator for Web Applications

An Interactive Client-Side Password Evaluation Interface

Technology Stack: HTML • CSS • JavaScript

2. ABSTRACT

Password Strength Checker is a web-based interactive application developed as a Full Stack Web Development use case assignment. The project provides a simple interface where users can enter a password and receive immediate visual feedback about its strength.

The application uses HTML for page structure, CSS for styling and the strength indicator, and JavaScript to evaluate the password dynamically as the user types.

The project demonstrates practical front-end concepts such as DOM manipulation, event handling, conditional logic, dynamic styling, and interactive user interface development.

3. INTRODUCTION

Modern websites are expected to provide immediate feedback and a smooth user experience. Password creation is one area where users benefit from clear information about password quality.

The Password Strength Checker focuses on a password field, a strength indicator bar, and a feedback message. The application operates on the client side and demonstrates the use of fundamental web technologies.

4. USE CASE / PROJECT CONCEPT

4.1 Project Concept

Password Strength Checker is an interactive password evaluation interface developed as a practical web development use case. The concept is to help users understand password strength while entering a password.

4.2 Target Users

- Website users creating accounts
- Students learning web development
- Users who want immediate password feedback
- Developers demonstrating JavaScript interaction

5. OBJECTIVES

- To develop an interactive password strength interface.
- To create a clean and user-friendly password input form.
- To update the strength indicator dynamically as the user types.
- To implement password strength evaluation using JavaScript.
- To provide clear feedback such as Weak, Medium, or Strong.
- To demonstrate practical application of HTML, CSS, and JavaScript concepts.

6. TECHNOLOGIES USED

Defines the structure of the password form, indicator and feedback message.

6. TECHNOLOGIES USED

6.1 HTML

- Password input field
- Strength indicator area
- Feedback message
- Page structure

6.2 CSS

- Colors
- Typography
- Spacing
- Input styling
- Indicator appearance
- Responsive presentation

6.3 JavaScript

- Reads password input
- Evaluates password strength
- Updates DOM elements
- Changes indicator dynamically

6.4 GitHub

GitHub is used to store and manage the project files. The project contains the main HTML, CSS and JavaScript files.

7. FEATURES & FUNCTIONALITIES

7.1 Password Input

Users can enter a password through a dedicated password field.

7.2 Real-Time Evaluation

JavaScript evaluates the password while the user types.

7.3 Strength Indicator

A visual bar represents the current password strength.

7.4 Feedback Message

The application displays a readable strength message.

7.5 Dynamic Update

The interface changes immediately without page refresh.

7. FEATURES & FUNCTIONALITIES

7.6 Password Rules

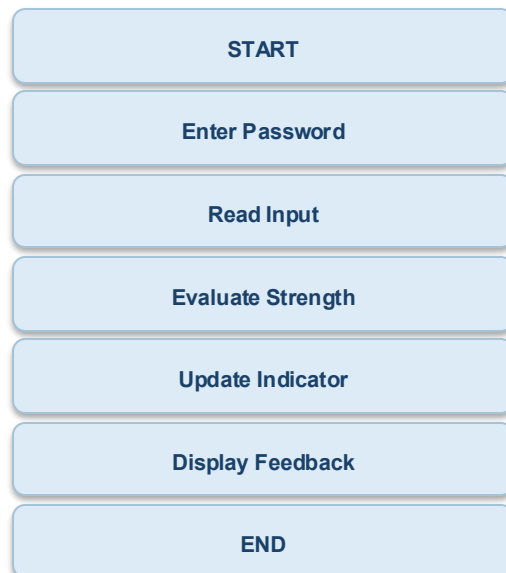
- Minimum length check
- Uppercase character check
- Number check
- Special character check
- Strength score based on matched rules

7.7 Strength Levels

- Weak – low score
- Medium – partial requirements
- Strong – all major requirements satisfied

8. SYSTEM DESIGN

Overall Workflow



9. IMPLEMENTATION

9.1 HTML Implementation — index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Password Strength</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>Password Strength</h1>
    <input type="password" id="password"
      placeholder="Enter password">
    <div class="strength-bar">
      <div id="strength"></div>
    </div>
    <p id="message">Enter a password</p>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

Purpose

HTML creates the page structure: password input, strength bar, feedback message, and connection to CSS and JavaScript.

9. IMPLEMENTATION

9.2 CSS Implementation — style.css

```
body {
  font-family: Arial, sans-serif;
  background: #f2f2f2;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
}

.container {
  width: 350px;
  padding: 30px;
  background: white;
  border-radius: 10px;
  text-align: center;
}

input {
  width: 100%;
  padding: 12px;
  box-sizing: border-box;
}

.strength-bar {
  width: 100%;
  height: 8px;
  background: #ddd;
  margin-top: 15px;
}

#strength {
  height: 100%;
  width: 0%;
}
```

Purpose

CSS controls the page layout, card appearance, input field, spacing, and strength indicator.

9. IMPLEMENTATION

9.3 JavaScript Implementation — script.js

```
const password = document.getElementById("password");
const strength = document.getElementById("strength");
const message = document.getElementById("message");

password.addEventListener("input", function () {
  let value = password.value;
  let score = 0;

  if (value.length >= 8) score++;
  if (/[A-Z]/.test(value)) score++;
  if (/[0-9]/.test(value)) score++;
  if (/[^A-Za-z0-9]/.test(value)) score++;
```

Purpose

JavaScript reads the password whenever the user types and checks length, uppercase letters, numbers, and special characters.

9. IMPLEMENTATION

9.3 JavaScript Implementation — script.js (continued)

```
if (value.length === 0) {  
    strength.style.width = "0%";  
    message.textContent = "Enter a password";  
} else if (score <= 1) {  
    strength.style.width = "25%";  
    message.textContent = "Weak";  
} else if (score <= 3) {  
    strength.style.width = "60%";  
    message.textContent = "Medium";  
} else {  
    strength.style.width = "100%";  
    message.textContent = "Strong";  
}  
});
```

Purpose

The score is converted into a visual strength level: Weak, Medium, or Strong. The bar width and message update immediately.

9. IMPLEMENTATION

9.4 Project Structure

PasswordStrengthChecker/

```
├── index.html
├── style.css
└── script.js
```

index.html

Page structure and password input

style.css

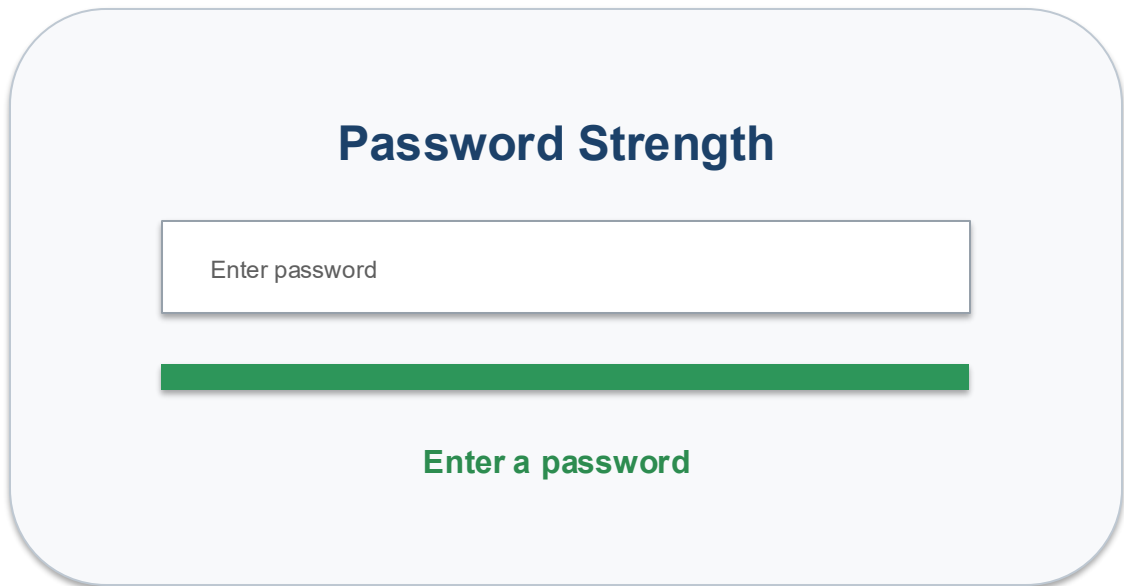
Layout and visual styling

script.js

Strength checking and dynamic updates

10. USER INTERFACE / SCREENSHOTS

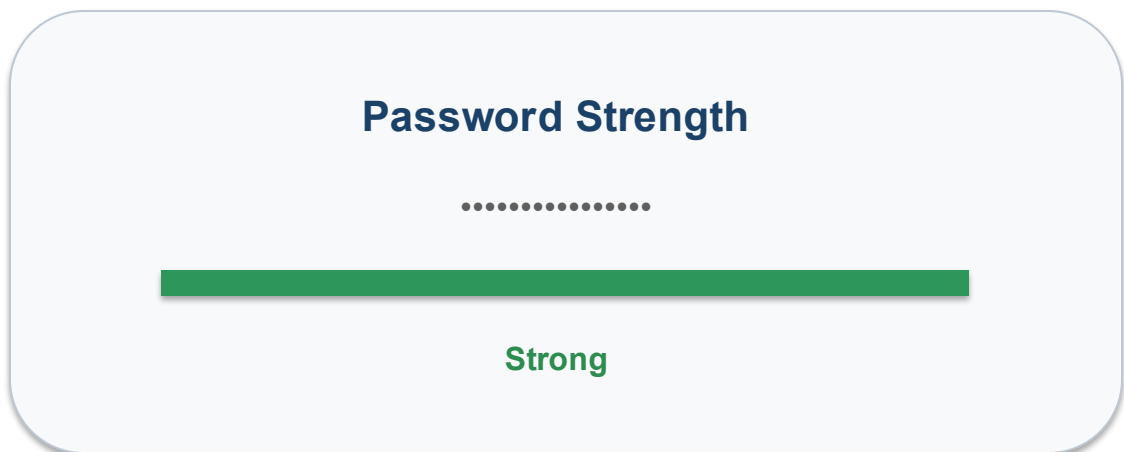
Screenshot 1 – Password Strength Page



A screenshot of a web form titled "Password Strength" in a dark blue font. Below the title is a white text input field with the placeholder text "Enter password". Underneath the input field is a solid green horizontal bar. At the bottom of the form, the text "Enter a password" is displayed in a green font.

Description: The home page contains the password field, strength indicator bar and feedback message.

Screenshot 2 – Strong Password Output



A screenshot of the same "Password Strength" form. The title "Password Strength" is at the top. Below it is a row of ten small grey dots. Underneath the dots is a solid green horizontal bar. At the bottom, the word "Strong" is displayed in a green font.

Description: The indicator reaches the strong level when the password satisfies the required conditions.

11. LIMITATIONS

The current version of Password Strength Checker is primarily a front-end web application and therefore has some limitations.

- It does not currently contain a backend server.
- Password data is not stored in a database.
- The strength rules are based on the implemented JavaScript logic.
- User authentication is not implemented.
- The project is intended as a demonstration of front-end concepts.

12. CONCLUSION

Password Strength Checker successfully demonstrates the development of an interactive web interface using HTML5, CSS3, and JavaScript. The project provides real-time password feedback through a strength indicator and dynamic message.

Through this project, important concepts of web development such as HTML structure, CSS styling, JavaScript event handling, DOM manipulation and interactive user interfaces were practically implemented. The project can later be extended with backend integration, secure authentication and stronger password policies.

13. REFERENCES

- MDN Web Docs – HTML, CSS and JavaScript documentation.
- W3Schools – HTML, CSS and JavaScript learning resources.
- GitHub Documentation – Repository management and project hosting.
- Visual Studio Code Documentation – Development environment and code editing resources.
- ECMAScript / JavaScript Documentation – JavaScript language features and functionality.
- ChatGPT – Assistance in understanding and developing the case study.

Case Study Reference

Use Case – 03: Design a form with a password field and a strength indicator bar that updates dynamically as the user types. Style the indicator with CSS and update it using JavaScript.

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